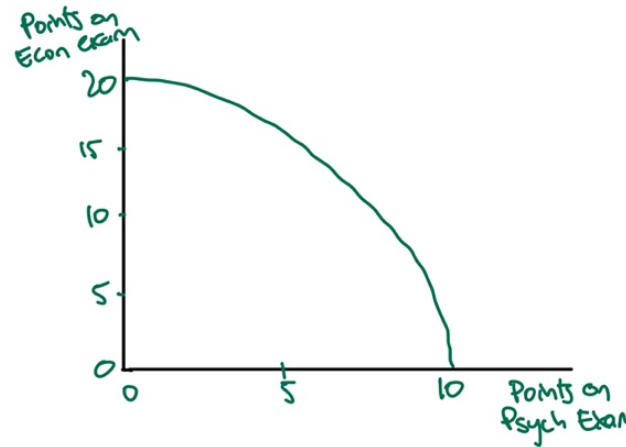


ECON 001 - Midterm 1 (version a)

September 28, 2022

There are 25 problems, each with 4 possible responses. Choose the single best response. Fill it in on the Scantron sheet. Marvel at how much you have learned!

- 1) A binding price ceiling
 - a) results in a lower quantity being sold
 - b) lowers the price
 - c) is above the equilibrium price
 - d) a and b
- 2) Bob dropped out of medical school. He must
 - a) believe he could make more money in some profession other than being a doctor.
 - b) believe the financial benefit of attending medical school is smaller than the cost.
 - c) think that the opportunity cost of continuing in medical school is too high.
 - d) be worried about having too much student loan debt.
- 3) Jhumpa was starving! Her WTP for a cookie was \$9. She bought a cookie at a store (7-Eleven) for \$3. Jhumpa's economic surplus from that transaction
 - a) is \$6.
 - b) depends on how much the store values the cookie.
 - c) depends on how you frame the problem.
 - d) b and c
- 4) The below production possibility frontier shows how many additional points Bob can get on his economics and



psychology exams based on how he spends his time studying. The PPF

- a) indicates the optimal number of hours for Bob to study for psychology.
 - b) shows the tradeoff Bob faces in studying psychology versus economics.
 - c) doesn't make any economic sense.
 - d) shows Bob needs to study for both classes.
- 5) The PPF (in the above problem) shows that

- a) as he gets more points on his economics exam, there is an increasing marginal benefit of points on the economics exam.
 - b) as he gets more points on his economics exam, there is an increasing marginal cost of points on the economics exam
 - c) as he gets more points on the psychology exam, there is a decreasing marginal cost of points on the psychology exam.
 - d) b and c
- 6) Tripp's favorite band was coming to town. Tripp knew the line to buy a \$20 ticket would take two hours, and the value of his time for waiting in line is \$25/h. Unfortunately, when he got to the front of the line, there were no more \$20 tickets: Only \$50 tickets (which Tripp to the \$20 tickets preferred) remained. Tripp, who always acts rationally, would
- a) Probably not buy the ticket since \$50 is a lot more than \$20.
 - b) Probably buy the ticket.
 - c) [We can't say what he'd do since we don't know enough about Tripp's WTP to attend the concert.]
 - d) [It depends on how much better the \$50 ticket is than the \$20 ticket.]
- 7) Suppose at the beginning of the semester, you paid \$10,000 for an unlimited meal plan at the cafeteria, which permits you to enter the cafeteria at any time to eat a meal. The marginal financial cost to you of eating a meal (that is, what it actually costs you in dollars) at the cafeteria is
- a) zero.
 - b) depends on how many times you eat at the cafeteria this semester.
 - c) depends on how much you like the food.
 - d) depends on how much a meal costs at an equally convenient location.
- 8) Applying the rational rule when deciding how many cookies to consume will
- a) lead you to the highest economic surplus.
 - b) cause you to eat cookies until your willingness to pay for a cookie is zero.
 - c) cause you to continue to eat cookies as long as the marginal cost is zero.
 - d) lead you to substitute cookies for milk when the price of milk increases.
- 9) At the ice cream shop near my house, milkshakes cost \$5, and I buy three. Last weekend, I was at the airport where milkshakes cost \$9 dollars, and I bought five. Assume the milkshakes at each location are equal size and quality. Then:
- a) Somewhat unusually, my demand curve for milkshakes slopes up.
 - b) My demand curve from milkshakes at the airport must be different than my demand curve for milkshakes at the ice cream shop near my house.
 - c) My consumption behavior is inconsistent and irrational.
 - d) To understand my behavior, we need to know about marginal cost of supply.
- 10) It's "Dollar Cone Day" at Ben & Jerry's Ice Cream Shop, a day when they lower the price for an ice cream cone to one dollar. On Dollar Cone Day a lot more ice cream cones are bought and sold. An explanation of that is:
- a) There's a shift in demand because, of the fall in price.
 - b) There's a movement along the demand curve.
 - c) The cost of supplying ice cream has fallen, which is why Ben & Jerry's lowered their price for a day.

- d) The ceteris paribus assumption.
- 11) As the quantity supplied increases, the marginal cost of production
- a) Is constant.
 - b) eventually decreases.
 - c) eventually increases.
 - d) always increases, even at low levels of supply.
- 12) In a perfectly competitive market,
- a) suppliers try to compete with one another by making improvements to their products.
 - b) there tends to be a few large sellers that have a lot of influence on the market.
 - c) there's a single price for the good.
 - d) All of the above.
- 13) Rebekah runs a cookie business. Each week, she rents a "cookie stand" (fixed cost). In addition to that cost, she also has the variable costs of ingredients, packaging, labor, etc. Suppose last week Rebekah rented a cookie stand for \$10 and she sold 15 batches of cookies. But this week the rental cost of the stand fell to \$5. (Suppose the demand for Rebekah's cookies and Rebekah's other costs are the same as the previous week.) Then
- a) Rebekah will sell more cookies this week, because now her costs are lower.
 - b) Rebekah will sell the same number of cookies this week as she did last week.
 - c) There is no effect on Rebekah's total profits this week, because the cookie stand is a sunk cost.
 - d) She will try to lower her price this week now that her total costs are lower.
- 14) Reasons that supply curves slope up include
- a) There are fewer suppliers willing to supply goods at higher price points.
 - b) The marginal product of inputs—how many extra goods an additional unit of input creates—eventually decline as a firm uses more of those inputs.
 - c) The marginal cost of inputs increases as a firm uses more inputs.
 - d) b and c.
- 15) Suppose in September Bobby paid \$200 to reserve a spot for a camping trip over Thanksgiving weekend (the \$200 is nonrefundable, but there would be no additional financial expenses associated with the trip). But yesterday an attractive classmate of Bobby's invited Bobby to come to their house for Thanksgiving (there would be no need for Bobby to spend any money if Bobby went to the classmate's house).
- a) The \$200 is a sunk cost.
 - b) Bobby compares his current WTP for going camping to his WTP for going to his classmate's house, and Bobby chooses the option for which he has the highest WTP.
 - c) If Bobby expects to have more than \$200 worth of enjoyment hanging out with this attractive classmate, Bobby will choose to go to the classmate's house.
 - d) a and b
- 16) Suppose the market for widgets in your town is perfectly competitive. Widget makers discover a new way of making widgets that requires fewer inputs. As a result,
- a) the widget makers are able to raise their prices.
 - b) the equilibrium price falls, and more widgets are sold.

- c) the equilibrium price stays the same but more widgets are sold.
 - d) demand for widgets increases.
- 17) When there is a shortage,
- a) the demanders most in need of the good are unable to get one.
 - b) market forces put upward pressure on the price.
 - c) the demand curve shifts.
 - d) it could be because the government has imposed a price floor.
- 18) Suppose pencils and notebooks are complements. Then when the price of notebooks falls,
- a) the equilibrium price of pencils increases.
 - b) the equilibrium price of pencils falls.
 - c) there is a shift in the supply curve of pencils.
 - d) the equilibrium quantity of pencils would stay the same.
- 19) As the number of reasonable substitutes for a good increases, *ceteris paribus*,
- a) demand for the good becomes more elastic.
 - b) demand for the good becomes less elastic.
 - c) shoppers for the good become less price sensitive.
 - d) a and c
- 20) As my income increases, *ceteris paribus*, I buy more cups of coffee. Thus:
- a) The income elasticity of demand for cups of coffee is positive but less than 1.
 - b) Cups of coffee are a luxury good for me.
 - c) Cups of coffee are a normal good for me.
 - d) Cups of coffee are complements to relaxing at a coffee shop.
- 21) When the government imposes a tax on a simple market of the type discussed in class,
- a) it should spread the economic burden of the tax by having suppliers pay half the tax and demanders pay half the tax.
 - b) the incidence of the tax depends only on which side the government requires to pay the tax.
 - c) the incidence of the tax depends on the relative elasticities of demand and supply of the good.
 - d) the quantity of goods bought and sold could increase.
- 22) Suppose your favorite coffee shop only accepts payment by credit card. Credit card companies charge the coffee shop 3 percent of every transaction (i.e., every time a customer pay the coffee shop, the coffee shop pays the credit card company 3 percent of the amount paid by the customer). Suppose credit card companies consider increasing the fee they charge the coffee shop from 3 percent of each transaction's value to 5 percent. If they did, then
- a) the economic surplus of both the coffee shop and its customers would fall.
 - b) the amount of coffee supplied would be the same, but the supplier's profits would fall.
 - c) the amount of coffee purchased would be the same, but the price the coffee shop charges customers would increase.
 - d) the total amount of fees collected by the credit card companies would increase.

- 23) Sebastian sells chocolate and wanted to figure out the price elasticity of demand of his customers. Almost every week, he charges a price of \$1.50 and sells 105 chocolate bars. But last week he charged a price of \$2.50 and sold 95 bars. Using that data and the midpoint formula, he computed the (arc) price elasticity of demand for his chocolate bars to be _____.

a) $\frac{10}{-1} = -10$

b) $\frac{\frac{10}{100}}{\frac{-1}{2}} = -0.2$

c) $\frac{\frac{10}{100}}{\frac{-1}{2}} = -0.2$

d) Not enough information is provided.

- 24) Suppose the demand function (equation for demand) is $Q_D = 12 - P$ and the supply function (equation for supply) is $Q_S = \frac{1}{2}P$. If the price is 2, then there is a _____ of _____ units.

a) shortage; 10

b) surplus; 1

c) surplus; 9

d) shortage; 9

- 25) Suppose the demand function (equation for demand) is $Q_D = 8 - P$ and the supply function (equation for supply) is $Q_S = \frac{1}{3}P$. If the government levied a tax of 4 dollars on sellers, then the new equilibrium quantity would be _____ unit(s).

a) 1

b) 2

c) 3

d) 6

Do not turn over this exam until instructed to. During the exam, you may use this page (or any part of this exam packet) for scratch paper. Good luck!